

Computer Science Engineering: Theory

Practice Paper

Q1. (Web Dev) Explain the difference between client-side and server-side rendering.

Why is responsive design crucial?

Q2. (OS) Differentiate between Paging and Segmentation. Explain how an OS handles process scheduling.

Q3. (OOP) Describe the four pillars of Object-Oriented Programming with a real-world analogy.

Q4. (DBMS) Define Normalization. Explain the difference between a Primary Key and a Foreign Key with an example.

Q5. (Data Structure) Compare Linked List and Array. Under what conditions would you prefer a Linked List over an Array?

Q6. (Networks) Explain the TCP 3-way handshake process. Why is TCP preferred over UDP for file transfers?